

FAREAL AHMED

Software Engineer

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SKILLS

Languages & Frameworks: Java, Kotlin, Spring Boot, JavaScript, TypeScript, React.js, Next.js, Node.js, Express.js

AI & LLM Tooling: LangChain4j, Ollama, pgvector, RAG pipelines, Claude API, MCP (Model Context Protocol), prompt engineering, vector embeddings, semantic search

APIs & Integration: RESTful APIs, GraphQL, OpenAPI/Swagger, WebSockets, MCP servers, webhook orchestration

Databases: PostgreSQL (pgvector), Microsoft SQL Server, MongoDB, Prisma, Hibernate, SQLite

Cloud & DevOps: AWS, GCP, Docker, Git, GitLab, CI/CD, Kibana, Grafana, DigitalOcean, Sentry, Umami

Development Practices: Agile (Scrum), TDD, Clean Architecture, Microservices, RESTful API Design, systems thinking

EXPERIENCE

Software Engineer | Compass Operations Pty Ltd., Melbourne | *June 2022 - September 2025*

- **Built end-to-end full-stack modules for Griddle, a school timetable and daily schedule management system, serving users across Australia, UK and Europe** (timetables, course selections, exams, daily-organisation) using Java, Kotlin, Spring Boot, React, and TypeScript.
- **Engineered scalable backend services** implementing RESTful and GraphQL APIs. Developed React/TypeScript UIs with modular, accessible components adhering to enterprise design systems.
- **Re-engineered asynchronous job engine** into hybrid sync/async processor using Factory pattern and Java Suppliers, enabling schools to make immediate exam changes that propagated instantly across timetables and daily organisation, while maintaining 100% code reuse.
- **Optimized system performance** by designing lean DTOs and efficient data-fetching strategies, reducing page load times by 40%.
- **Implemented recursive backtracking algorithm** to resolve 30+ edge cases in data selection, eliminating scheduling conflicts and ensuring 100% data integrity during critical course selection time. Refactored legacy service layers using Clean Architecture principles, reducing technical debt.
- **Maintained comprehensive test suites (TDD)** and monitored CI/CD pipelines in GitLab, ensuring stable releases aligned with academic deadlines.

Technical Trainer | BASIS Institute of Technology & Management, Dhaka | *February 2013 - October 2014*

- Trained 3 cohorts in C#, .NET, OOP, and clean coding practices; developed curriculum based on industry standards.

Software Engineer | ADDIE Soft Ltd., Dhaka, Bangladesh | *August 2011 - January 2013*

- Built core features for Oddhoyon, a school management system used by 1000+ schools in Bangladesh.
- Led migration from desktop (Windows Forms) to ASP.NET MVC, improving scalability and maintainability.

PERSONAL PROJECTS

TrolleyToday — Grocery Price Comparison App for Australia (Active Development)

- Built and deployed a mobile-first PWA that lets Australian shoppers compare grocery prices across Woolworths, Coles, IGA, Aldi, and specialty grocers from a single interface, with cheapest/most-expensive items auto-highlighted and price-per-unit normalisation (\$/kg, \$/L) for fair comparison.
- Designed an agentic AI assistant using tool-calling architecture — the assistant autonomously decides whether to run a semantic product search, batch-price multiple items, or invoke a two-step recipe pipeline (LLM ingredient extraction → parallel vector search per ingredient). The primary model is Llama 3.3-70b via Groq with automatic transparent fallback to Gemini 2.0 Flash on rate-limit errors.
- Built hybrid search combining LLM query expansion with local BGE-small-en-v1.5 embeddings via Xenova/Transformers and pgvector, fused with PostgreSQL keyword matching and pg_trgm fuzzy matching for typo tolerance. Responses stream token-by-token via ReadableStream.
- Implemented a strict layered architecture (UI → hooks → services → API routes → data layer) with TypeScript contracts at every boundary, keeping components fully decoupled from data access. Built offline-capable shopping list using Zustand + localStorage with two view modes (by-product and by-store), quantity steppers, and one-tap PDF export (jsPDF) grouped by store with subtotals.

- Deployed on DigitalOcean with Docker-managed PostgreSQL 16, IP-based rate limiting, Sentry error tracking, and self-hosted Umami analytics.
- Tech: Next.js 14, TypeScript (strict), Tailwind CSS, PostgreSQL 16 (pgvector, pg_trgm), Zustand, Groq API, Google Gemini API, Xenova/Transformers, jsPDF, Sentry, Umami, Docker, DigitalOcean
- Live at <https://trolleytoday.com>

Doc Whisperer — AI-Powered Document Assistant (RAG Application)

- Built a full-stack RAG application that lets users upload PDF and DOCX files and chat with them using natural language. The backend uses Spring Boot with Java 21, LangChain4j for orchestrating the retrieval pipeline, and Apache Tika for document parsing. Text is chunked, embedded using all-MiniLM-L6-v2 (384-dimensional vectors), and stored in PostgreSQL with pgvector for cosine similarity search. Local LLM inference runs through Ollama (llama3), keeping all data on-machine with no external API calls. The frontend is React 18 with TypeScript, Tailwind CSS, and Shadcn/UI components. The full RAG pipeline — upload, chunk, embed, retrieve, generate — runs end-to-end with OpenAPI/Swagger documentation for every endpoint.
<https://github.com/farealahmed/doc-whisperer>

Video Vault

- Built end-to-end video streaming application integrating Spring Boot REST backend with React frontend. Designed PostgreSQL database schema using Docker, implemented secure authentication with Spring Security, and engineered efficient multipart file handling for large video uploads.
<https://github.com/farealahmed/my-video-vault>

EDUCATION

BSc in Computer Science and Engineering

Ahsanullah University of Science and Technology, Dhaka, Bangladesh | 2007 - 2011